

MICROCYTIC ANEMIA AND CHANGES IN FERROKINETICS AS LATE AFTER-EFFECTS OF GLUCAN ADMINISTRATION IN MURINE HEPATITIS VIRUS-INFECTED C57BL/10ScSnPh MICE

J. VÁCHA,*† V. ZNOJIL,* M. POSPÍŠIL,‡ J. HOLÁ* and I. PIPALOVÁ*

*Institute of Pathological Physiology, Medical Faculty, Masaryk University, Komenského nám. 2, 662 43 Brno, Czech Republic; and †Institute of Biophysics, Academy of Sciences of the Czech Republic, 612 65 Brno, Královopolská 135, Czech Republic

(Received 1 July 1991 and in final form 16 July 1993)

Abstract — Mild microcytic anemia (without changes in mean corpuscular hemoglobin concentration, MCHC) was discovered 6–14 weeks after a single s.c. administration of 4 mg of particulate glucan to C57BL/10ScSnPh mice serologically positive for murine hepatitis virus (MHV). The anemia was associated with granulocytosis, decreased body weight and spleen hypertrophy. The overall intensity of erythropoiesis was measured by ⁵⁹Fe-incorporation into the heme of erythropoietic organs. The localization of erythropoiesis became markedly redistributed — heme production was suppressed in the bone marrow while a several-fold increase was recorded for the spleen. A new steady state was also discovered in ferrokinetics: an iron pool localized away from the blood, erythropoietic organs and the liver was significantly elevated, and hypoferremia was detected. Anemia and wasting of mice were not observed in the same mouse strain free of MHV.

A single administration of particulate glucan resulted in late impairment of red blood cell formation in the C57BL/10ScSnPh mouse strain infected with the mouse hepatitis virus. The anemia shares a number of features with those observed for the anemia of chronic diseases.

Glucan isolated from the cell walls of *Saccharomyces cerevisiae* (β -(1→3)-D-glucan) has been widely studied as a potent stimulant of reticuloendothelial function and as a broad spectrum enhancer of host defense mechanisms against bacterial, viral, fungal, parasitic infections, and neoplasia (DiLuzio, 1983, 1985). Glucan administration to mice is followed by an increased production in several hemopoietic compartments (Patchen & MacVittie, 1983). Increased proliferation of the granulocyte–macrophage colony-forming cell (CFU-GM) following glucan administration is associated with its ability to induce the production or release of granulocyte–macrophage colony stimulating activity (Patchen, 1983). The hemopoietic effects of glucan manifest themselves through a stimulation of the production of the stem cells of hemopoiesis and granulocytogenesis, and have been exploited in enhancing hemopoietic recovery and in increasing the radiation resistance of gamma-irradiated mice

(Patchen, 1983; Patchen, D'Alesandro, Brook, Blakely & MacVittie, 1987; Pospíšil, Hofer, Pipalová, Viklická, Netíková & Šandula, 1992; Pospíšil, Jarý, Netíková & Marek, 1982).

On the other hand, however, we have found that the effects of glucan may not only be beneficial, but under certain circumstances they can also become unfavorable even to lead to wasting and premature deaths of the animals. A follow-up study of glucan effects carried out for a number of weeks after its administration (Pospíšil *et al.*, 1988) revealed this effect in the mouse strain C57BL/10ScSnPh produced in our institute's own farm. In this study, we decided to analyze more closely the mechanism of the origin of the anemia in these animals by performing a detailed ferrokinetic study. After completing this task we have learned, however, that the animals used had enhanced plasma titers of the murine hepatitis virus (MHV). Therefore, we added some experiments to check whether the wasting and

†Author to whom correspondence should be addressed.

the anemia observed in these mice after glucan administration was co-conditioned by their MHV positivity. It turned out, indeed, that while the MHV infected animals showed profound changes in their body weight, blood cell indices and ferrokinetics after glucan, the MHV-free animals displayed only some ferrokinetics deviations after glucan compensated completely in regard to erythropoiesis, without any signs of wasting.

EXPERIMENTAL PROCEDURES

Animals

MHV mice. Males of the inbred C57BL/10ScSnPh mouse strain were used, produced in conventional manner by the institute's own farm as category I animals [classification according to Bondy, Klír, Příbylová, Svoboda, Jelen & Pravenec (1987)]. After completing the experiments, significantly enhanced titers for MHV were found using complement-fixation reaction in virtually all tested animals of the same strain and origin (titers having been in the range of 1 : 8–1 : 32, tests having been performed in the reference laboratory of the Institute of Physiology, Czechoslovak Academy of Sciences, Prague). *Pasteurella pneumotropica* was absent. The animals were 3–4 months old, weighing 27–32 g at the start of the experiments. They were kept in standard conditions with natural lighting regime and a temperature of $22 \pm 2^\circ\text{C}$. Sterilized pellet diet (DOS — 2/St Velaz) and HCl-treated water (pH 2–3) were given *ad libitum*. Animals were kept in plastic cages in groups of 20. Experimental and control treatments were compared for one and the same cage. (The term “experimental” animal means a glucan-treated mouse in the present paper, “control” animal means an animal without glucan, irrespective of MHV status.)

MHV-free mice. These were produced in the same institute's facility, after necessary sanitary measures had been taken in the rooms used for the C57BL/10ScSnPh breeding. The new parent mice were obtained from the same source as before [Institute of Experimental Biology and Genetics, Czechoslovak Academy of Sciences, Prague — Krč, abbreviation *Ph*, see Staats (1981)] and were repeatedly serologically MHV tested with negative results. All other conditions of breeding and handling were the same as for the MHV-infected mice. The animals were of comparable age and body weight as those of MHV mice at the start of the experiments.

Materials

Glucan (in particulate form) was isolated from the cell walls of *S. cerevisiae* by a modification of the methods of Hassid, Joslyn and McCready (1941), and Suzuki, Yokota and Sakaguchi (1967) using alkaline–acid hydrolysis (the isolation was performed by Dr J. Jarý, Institute of Chemical Technology, Prague). The glucan preparation was suspended in sterile saline and injected in a single s.c. dose of 4 mg/mouse at a volume of 0.5 ml. Controls were injected with the same volume of saline. The glucan suspension forms a heterogeneous system of variously sized particles, the largest of which being about 20 nm in diameter. Endotoxin contamination was less than $1 \mu\text{g}/1 \text{ mg}$ as determined by the Limulus amoebocyte lysate assay (Dialab).

Routine hematological techniques

MHV mice. Blood was withdrawn from a fine incision in the tail vein in approximate volumes of 0.02 ml, repeatedly from the same animals (10 experimental and 10 control mice at the start of the experiment) on days 0, 8, 15, 22, 29, 43, 57 and 85 after glucan administration. Leukocyte and erythrocyte counts were determined with a Coulter Counter (model ZF, Coulter Electronics Ltd, U.K.), the mean red cell volume (MCV) and hematocrit with a Coulter MCV/Hct Accessory, and hemoglobin concentration by means of a Hemoglobinometer Coulter. Granulocytes and lymphocytes were calculated on the basis of blood smear differentiation. The total volume of circulating red cell mass was determined in a separate group of mice (10 experimental and 10 control animals) using a dilution of ^{59}Fe -labelled erythrocytes [according to Vácha (1975)] on days 50 and 79 after glucan administration.

MHV-free mice. The peripheral blood parameters were measured in a group of 10 control and 10 experimental mice on days 0, 14, 28, 42 and 56 after glucan administration; only the last two intervals have been evaluated in the present paper.

Ferrokinetics

MHV mice. On days 41 and 70 after glucan administration a separate group of animals (5 experimental and 5 control mice for each experiment) was injected i.p. with 0.5 ml volumes of ^{59}Fe bound to transferrin according to our method (Holá, Vácha & Boháček, 1979), i.e. by incubating ^{59}Fe -citrate (ROTOP, GDR, specific activity 284 GBq/g Fe) with isogenic serum in a 1 : 2 volume ratio, and removing the unbound ^{59}Fe by means of Dowex-1 X-

Table 1. Indices of hematopoiesis in peripheral blood (MHV mice)

	Control		Glucan		Percent		P
	Mean	S.E.	Mean	S.E.	Mean	S.E.	
Erythrocytes (10^6 per μ l)	10.17	0.14	10.15	0.20	99.8	2.4	n.s.
Hematocrit (%)	46.77	0.70	40.07	0.75	85.6	2.1	<<0.001
Hemoglobin (g/l)	148.5	2.6	124.7	2.3	84.0	2.1	<<0.001
MCV (μ m ³)	46.67	0.27	40.66	0.30	87.1	0.8	<<0.001
MCH (pg)	14.82	0.20	12.46	0.23	84.0	1.9	<<0.001
MCHC (%)	31.75	0.47	30.64	0.61	96.5	2.4	n.s.
Total RBC mass							
volume (ml)*	0.79	0.04	0.72	0.04	91.1	6.9	n.s.
Leukocytes (10^3 per μ l)	13.98	2.21	17.83	1.52	127.5	22.9	n.s.
Lymphocytes (10^3 per μ l)	10.65	1.38	9.06	1.82	85.1	20.3	n.s.
Granulocytes (10^3 per μ l)	3.33	1.12	8.77	0.88	263.4	92.4	<0.001

*50th and 79th day.

4 ion exchanger, mesh 100–200, of Fluka. Six hours later the animals were bled under ether anesthesia, the blood was transferred on heparin powder, centrifuged, erythrocytes were washed and the radioactivity in 100 μ l of erythrocytes and 100 μ l of plasma was measured. Radioactivity was measured in the gamma spectrophotometer Compu Gamma 1282 (LKB) in the range of 1.0–1.3 MeV; the measuring period was set so that the measurement error should not exceed 1%. Background count rate was subtracted from the measured value for the samples and the resulting value was converted to the percentage of a ^{59}Fe standard measured in parallel.

The spleen was wet-weighed, the liver was washed with saline *in situ* via v. portae, wet-weighed, and aliquots of the tissue were taken for radioactivity counting. Cleared skeletons were obtained by way of necrophagous activity of the larvae and imagoes of the *Dermestes vulpinus* Fbr. beetle kept according to Grulich (1977). After total radioactivity counting the skeletons were ground at a temperature of liquid nitrogen and heme iron was extracted from them according to our original method (Vácha, Dungal & Kleinwächter, 1978). ^{59}Fe contents in the skeleton and the spleen were numerically reduced by a value corresponding to the radioactivity of the plasma and the erythrocytes circulating in these organs according to Vácha, Holá, Dungal and Znojil (1980). The correction is based on assuming that the volume of the circulating plasma and the erythrocytes in the bones and spleen was not influenced by glucan administration.

In another group of animals (10 experimental and 10 control animals) that were not given ^{59}Fe , the plasma Fe concentrations were measured on days

39 and 70 employing the method and the kit of C.F. Boehringer Mannheim (Biochemica test Combination, Eisen Farb-test, bathophenanthroline disulphonate), appropriately modified for the purpose of microanalysis. Registration spectrophotometer DU-8 (Beckman) was used to register absorption spectra in the wavelength range of 400–700 nm.

MHV-free mice. In a separate group of animals (14 experimental and 14 control animals) the radioactivity of ^{59}Fe was measured in the plasma and the circulating erythrocytes for each individual animal on day 52 after glucan administration in 14 consecutive time intervals from 10 min to 6 h following the administration of ^{59}Fe -transferrin complex. Curves of ^{59}Fe clearance from the plasma were obtained by an approximation of these values using the least square method.

Data processing. Calculations were based on the ratios of experimental to control values (in %, with corresponding standard errors), differences were evaluated by the unpaired Student's *t*-test.

RESULTS

Indices of hematopoiesis in the peripheral blood

MHV mice. Indices of hematopoiesis in peripheral blood obtained for days 43, 57 and 85 after glucan administration were not dependent on the time period and, therefore, their values could be pooled (Table 1). While the blood erythrocyte (RBC) concentration displayed no change after glucan administration, their mean corpuscular volume (MCV) was highly significantly reduced down to $87 \pm 1\%$ along with a corresponding reduction in mean corpuscular hemoglobin (MCH); mean

Table 2. Indices of hematopoiesis in peripheral blood (MHV-free mice)

	Glucan/control (%)		
	Mean	S.E.	P
Erythrocytes	101.1	3.2	0.73580
Hematocrit	96.7	3.3	0.34129
Hemoglobin	104.6	2.5	0.08280
MCV	99.0	1.3	0.41961
MCH	103.4	2.5	0.18954
MCHC	104.4	2.8	0.13402
Leukocytes	111.3	12.9	0.39287
Body weight	101.5	3.1	0.63101

Table 3. Indices of ferrokinetics (MHV mice)

	Control		Glucan		Percent		P
	Mean	S.E.	Mean	S.E.	Mean	S.E.	
Body weight (g)	34.4	1.3	28.2	0.8	82.8	4.4	<0.005
Spleen weight (mg)	88.2	2.5	152.8	10.3	174.2	13.9	<<0.001
Liver weight (mg)	1366	35	1282	40	94.0	3.9	n.s.
Plasma ^{59}Fe (per 100 μl)*	0.573	0.057	0.434	0.081	77.1	12.6	n.s.
RBC ^{59}Fe (per 100 μl)*	0.653	0.047	0.985	0.142	150.9	24.2	<0.1
Spleen heme- ^{59}Fe *	0.351	0.081	1.872	0.294	548	140	<0.002
Skeleton heme- ^{59}Fe *	12.630	0.878	5.651	0.804	44.6	6.7	<<0.001
Perfused liver ^{59}Fe *	3.81	0.30	2.15	0.34	56.6	9.9	<0.01
All organs ^{59}Fe *†	43.3	1.8	31.2	2.6	71.9	6.3	<0.01
Plasma Fe concentration $\mu\text{mol/l}$ ‡	35.5	3.6	<23.9	2.2	<67.2	8.5	<0.02

*Percent of radioactivity administered.

†Sum of all circulating blood, skeleton, spleen and liver.

‡39th and 70th day.

corpuscular hemoglobin concentration (MCHC) was almost unchanged. Independent measurements of circulating erythrocyte mass volume revealed that the reduction virtually corresponded to that of the mean volume of an individual cell, thus implying that microcytosis was virtually the only sign of anemia with unchanged hemoglobin density per unit erythrocyte volume, and no changes in the number of circulating erythrocytes and in plasma volume. A marked increase was recorded for granulocyte concentration (by a factor of about 2.5) while the total leukocyte concentration was increased only insignificantly. There was no significant change in the number of monocytes (not shown in Table 1).

MHV-free mice. Indices of hematopoiesis in the peripheral blood obtained for days 42 and 56 after glucan administration were not dependent on the time period and, therefore, their values could be pooled. Only relative values (in % of controls) are given in Table 2; obviously, the administration of glucan had not influenced the parameters of the

peripheral blood and the body weight in MHV-free mice within the period studied.

Indices of ferrokinetics

MHV mice. Indices of ferrokinetics obtained for days 41 and 70 after glucan administration did not differ from one another significantly and they were also pooled (Table 3). With a lower body weight of the experimental (= glucan-administered) mice the weight of the spleen increased markedly while that of the liver was reduced (along with the intensity of ^{59}Fe incorporation into the liver). The increase in ^{59}Fe incorporation into the circulating RBC was not significant while that into the heme fraction of the spleen was markedly elevated. ^{59}Fe incorporation into the heme fraction of the whole skeleton was reduced, and so was that into the non-heme fraction of the skeleton (equaling the difference between ^{59}Fe contents before extracting the heme fraction and the ^{59}Fe contained in the heme fraction — not shown in Table 3), amounting to $47.8 \pm 13.5\%$ ($P < 0.1$) and statistically indistinguishable from the reduction

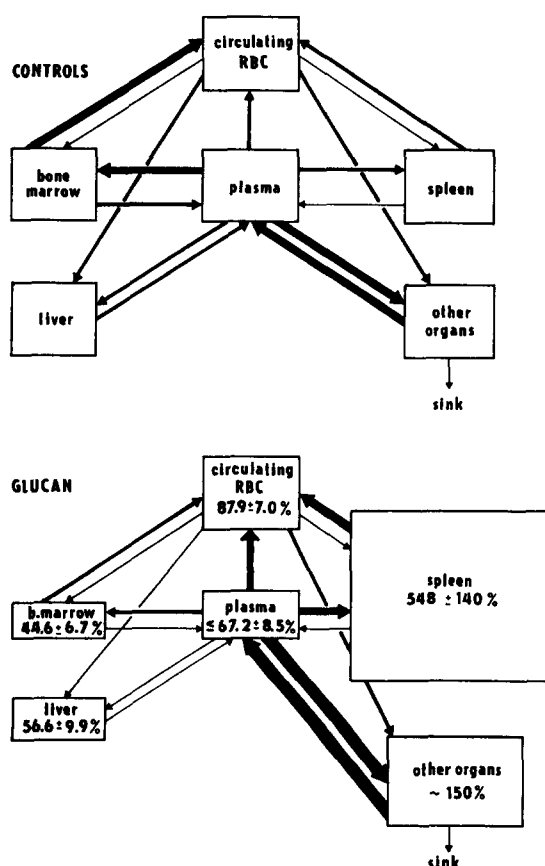


Fig. 1. Changes in the size of iron compartments and in flows between them in the 6th to 14th week after a single administration of 4 mg glucan in MHV-positive C57BL/10ScSnPh mice. The size of flows under physiological conditions are from Vácha, Znojil, Holá and Dungel (1982). The size of the compartments after glucan is given in percent of controls, the size of iron flows is represented by the thickness of the arrows (approximately).

in ^{59}Fe incorporation into the heme fraction. Plasma ^{59}Fe concentration in the experimental animals was reduced by about 1/4, and that of inactive Fe by at least 1/3. No mice died during the experiments.

By multiplying the Fe flow rate constants determined for experimentally intact animals (Vácha, Znojil, Holá & Dungel, 1982) by the relative sizes of the Fe compartments influenced by glucan, it was also possible to estimate the magnitude of the flows between the compartments in the experimental animals, which is illustrated in Fig. 1 as arrows of varying thickness.

A spectrophotometric examination of absorption by samples containing plasma iron complexed with batho-phenanthroline revealed the existence of a more or less

absorbing component(s) in the given spectrum range for the glucan-administered animals especially in an earlier period (day 39 after glucan administration). The presence of this component(s) fluctuated among individuals, which prevented its standard subtraction. The values of the plasmatic Fe concentration given in the bottom line of Table 3 were read in the maximum (535 nm) without correcting for the presence of this superposed moiety. The determination of the plasmatic Fe concentration is thus biased with a systematic error which led to a certain obliteration of the differences between experimental and control samples.

MHV-free mice. The half-time of ^{59}Fe clearance from the plasma of the experimental animals (= after glucan administration) was 63 ± 12 min as compared to 111 ± 26 min in the controls. However, the more pronounced reflux (indicated by a slower descent of the curve 2.5 h after ^{59}Fe administration) resulted in the fact that the total area under the curve remained unchanged for the experimental animals ($100 \pm 5\%$ of controls). No mice died during the experiments.

DISCUSSION

(1) Hemopoiesis after glucan administration Incorporation of ^{59}Fe into erythrocytes

Because of the decline in the total amount of plasma iron in the experimental (= glucan-treated) MHV animals, the plasma ratio of $^{59}\text{Fe}/\text{Fe}$ is enhanced during the whole period studied, and initially to $100 + (<67.2) = >150\%$ of control values, after 6 h to about $77.1 + (<67.2) = >115\%$ of control values. This rise implies a relative enhancement of the tracer accessibility (regarding to non-radioactive Fe) for the currently forming and young circulating erythrocytes. It is apparent, therefore, that the amount of ^{59}Fe in a certain volume of erythrocytes 6 h after ^{59}Fe administration must rise in the same ratio (MCHC being virtually unchanged); this is in accordance with the actually measured value $150.9 \pm 24.2\%$. This seemingly paradoxically high value is explained in this way.

Generation of a new Fe pool. An unexpected finding following glucan administration in the MHV mice is the generation (or a substantial expansion) of Fe pool localized in a site other than the blood, erythropoietic organs and the liver, the existence of which became apparent from the decrease in the total amount of ^{59}Fe in these tissues 6 h after its administration. As follows from the last but one line in Table 3, the magnitude of this pool amounts to $(100 - 43.3 \pm 1.8) = 56.7 \pm 1.8\%$ of the amount administered in the control MHV animals. In the glucan-treated MHV animals, the magnitude of this pool amounts to $68.8 \pm 2.6\%$ of control values. It is this expansion that makes the sum of the flows of

iron from plasma rise in the experimental animals, which entails the mentioned decline of plasma iron concentration (to $<67.2 \pm 8.5\%$); this lowering implies a decline in the accessibility of ^{59}Fe for the forming erythrocytes in erythropoietic organs.

It is not likely that a substantial part of this pool is located in the system of mononuclear phagocytes (MPS), although inflammatory macrophages are known to have an increased capacity to remove Fe from its complex with transferrin, i.e. apparently also from the plasma (Birgegard & Caro, 1984). As the organs under observation contain the far greatest proportion of MPS (Lynch, Lipschitz, Bothwell & Charlton, 1974), the new Fe pool would have to be observed as an increased incorporation of ^{59}Fe into their non-heme fraction, which did not happen: ^{59}Fe incorporation into the non-heme fraction of the bone marrow was decreased to a similar extent as that into the heme fraction, and the incorporation of ^{59}Fe into the washed (i.e. practically heme Fe-depleted) liver decreased to nearly a half the control value. (A possibility cannot, however, be excluded that the decrease in the concentration of non-labelled Fe in the plasma is partly the result of a higher retention by the MPS of the Fe leaking from disintegrating RBC, but no conclusion can be derived from the present work which did not concentrate on the question of Fe stores.) We could hypothesize that the newly generated pool represents an extension of an iron compartment interpreted in our ferrokinetic models (Vácha *et al.*, 1982, 1984) as the Fe contained in the extravascular space.

The relationship of the described anomalies in ferrokinetics to the ability of interleukin 1 (Klasing, 1984) in lowering plasma iron levels is not clear. An iron flow chart illustrating general changes in ferrokinetics following glucan administration is presented semiquantitatively in Fig. 1.

Intensity of hemoglobin production in erythropoietic organs. The liver devoid of circulating blood can serve as an indicator of the total accessibility of iron in organs in the time interval of 0–6 h after the administration of ^{59}Fe . The incorporation of the tracer fell to $57 \pm 10\%$ in this organ in the experimental MHV mice. [A possible retaining of Fe in the liver cells plays no important role in these calculations, as the half-life of the return of Fe from that storage pool equals 1.6 days even under physiological conditions, which is much more than 6 h (Vácha *et al.*, 1982).]

If we use the values of ^{59}Fe incorporation into the erythropoietic organs, the total heme production (in the bone marrow and the spleen) makes $\{(5.65 \pm 0.80) + (1.87 \pm 0.29)\} \div (12.63 \pm 0.88 + 0.35 \pm 0.08) = 58 \pm 8\%$. If we correct this value for the lowered accessibility of ^{59}Fe for maturing erythroblasts, i.e. by means of the ^{59}Fe incorporation value ascertained for the liver, we get $(58 \pm 8) \div (57 \pm 10)\% = 102 \pm 25\%$.

Under physiological conditions 87.3% of the total iron contained in erythrocytes become incorporated in the erythropoietic organs of C57BL/10ScSnPh mice, but the remaining 12.7% is incorporated directly from the plasma only after their leaving these organs (Vácha *et al.*, 1982). We could deduce that this directly incorporated quantity will even grow under the conditions of diminished accessibility of iron for the erythroblasts in the blood forming organs.

The quantity of hemoglobin in the peripheral blood equals the product of the hemoglobin production and the mean life span of erythrocytes (under the assumption that the steady state has been attained after the experimental interference with the variables mentioned). From the hemoglobin concentration of $84 \pm 2\%$ [possibly from the total erythrocyte mass volume and MHC, i.e. $(91.1 \pm 6.9) \times (96.5 \pm 2.4) = 88.0 \pm 7.0$] and the value of production of $102 \pm 25\%$, it follows that the mean life span of erythrocytes in the experimental MHV mice is $82 \pm 20\%$ of the controls. Regarding the variances given, one cannot decide whether it is the diminished production of erythrocytes and/or their diminished life span that accounts for the fall of hemoglobin or hematocrit values.

The ratio between the erythropoiesis in the spleen and in the bone marrow, and ferrokinetic parameters as sensitive indicators of the state of erythropoiesis.

A state known from a whole array of experimental conditions in mice manifests itself as a changed ratio between the intensity of ^{59}Fe incorporation into the bone marrow and the spleen. In this state, erythropoiesis is shifted into the spleen under its overall stimulation, sometimes even if erythropoiesis is suppressed in the bone marrow at the same time (as a result of depression or competition with other hemopoietic lines). Changes of ferrokinetics represent much more sensitive indicators of the changes in the erythropoietic system than the peripheral blood picture (Vácha *et al.*, 1988; Vácha, Znojil, Seidel & Barthel, 1990).

The results gained with the MHV-free mice after treatment with glucan conform well with this pattern. Experimental data testify for changes in ferrokinetics, without any manifestations in the blood picture; in spite of some changes in the metabolism of iron being elicited, it is not enough for causing any disturbances in erythropoiesis. The observed changes in the clearance of ^{59}Fe suggest that a new iron pool forms/expands under the influence of glucan even in MHV-free mice.

The striking increase in erythropoietic activity observed in the spleen may be a mechanism compensating for the insufficiency of the bone marrow in this respect — resembling a number of other well-documented cases of reduced intensity of erythropoiesis in the bone marrow. The whole situation is also paralleled at the level of progenitor cells (shortly after glucan administration): Patchen and MacVittie (1983) recorded a decrease in the concentration

of bone marrow colony-forming units-erythroid (CFU-E) and burst-forming units-erythroid (BFU-E) under glucan influence along with increased concentrations of these cell types in the spleen. The spleen being primarily engaged in erythropoiesis (Thomas, 1971), and in our experimental conditions being under the stimulatory influence of anemia, does not have its erythropoiesis suppressed as seen in the bone marrow, but, on the contrary, a compensation increase becomes predominant.

(2) Mechanism of anemia after glucan administration

The mechanisms of anemia studied here may be rather complex. It can be speculated that interleukin 1 is involved in the reduction of erythropoiesis in the bone marrow, as its production could be triggered experimentally by glucan *in vivo* (Sherwood, Williams, McNamee, Jones, Browder & DiLuzio, 1987), and this cytokine can directly inhibit erythropoiesis *in vitro* in concentrations corresponding to chronic inflammatory diseases (Maury, Andersson, Teppo, Partanen & Juvonen, 1988).

Also, increased production of tumor necrosis factor, which is known to occur after particulate glucan treatment (Sherwood, Williams & DiLuzio, 1986), could participate. Tumor necrosis factor can exhibit inhibitory effects on the growth of erythroid progenitor cells (Roodman, 1987). Faquin, Schneider and Goldberg (1992) have recently shown that inflammatory cytokines, including interleukin 1 and tumor necrosis factor, suppress hypoxia-induced erythropoietin production in Hep3B cells *in vitro*. Data suggesting the potential role of these inflammatory cytokines in the pathogenesis of the anemia of chronic diseases were summarized recently by Means and Krantz (1992).

Former experiments of Fruhman (1966) showed a decline in marrow erythropoiesis in mice after zymosan, a crude *S. cerevisiae* cell wall fraction, as well as after endotoxin (Fruhman, 1967). Thus, the role of endotoxin contamination of the particulate glucan used in our experiments cannot be excluded. It should be mentioned that particulate glucan can enhance the endotoxin sensitivity of animals (Bowers, Patchen, MacVittie, Hirsch & Fink, 1986).

Moreover, it cannot be excluded that the reduced erythropoiesis in the bone marrow is simply the result of a competition for a common stem cell of hemopoiesis (Hellman & Grate, 1967) under conditions favoring the white cell lineage (in our experiments this stimulation is documented by the peripheral granulocyte values). In the bone marrow, which is primarily engaged in myeloid production (Thomas, 1971), the outcome of this competition

may be that of reducing the determination of the cells into the red cell line.

The syndrome elicited by glucan in our experiments bears certain analogy to the anemia of chronic diseases [anemia of the ACD type, for review see, e.g. Lee (1983), Means & Krantz (1992)], although this aspect can be discussed only to a limited extent, as we have not investigated Fe stores in this work. In both cases hypoferrremia is observed as well as mild anemia along with reduced hematocrit, MCV and MCH values. Analogy can also be found in the reduced erythropoiesis in the bone marrow being counterbalanced by its increase in the spleen [in the mouse DeGowin & Lass (1985)], and in an increased granulocyte concentration in the peripheral blood. Some analogy cannot be ruled out between the appearance of new proteins in the plasma in ACD (Lee, 1983) and our observed anomaly in plasma absorption spectrum between 400 and 700 nm. On the other hand, however, we have not observed increased liver weight after glucan, which is associated in ACD with a synthesis of acute phase proteins (Lee, 1983).

(3) Genetically conditioned differences in immune reactivity; mice of C57BL origin as especially reactive

It remains to be ascertained whether the genetically conditioned reactivity of the mouse strain used contributes to the effects observed in our experiments. Recently, Gallin, Green and Patchen (1992) have described different responsiveness of C3H/HeN and C3H/HeJ mice to particulate and soluble glucans. Generally, mouse strain C57BL/10 and strains closely related to it seem to be high responders to various stimuli and immunomodulators. It has been established that the sensitivity of granulocyte and macrophage colony-forming cells (CFU-GM) to humoral factors enhancing the formation of corresponding colonies varies from one mouse strain to another, the colony-forming cells of the strain C57BL being regarded as particularly responsive (McNeill & Fleming, 1973). The inflammatory reaction to latex particles and *Listeria monocytogenes* extract expressed as the number of monocytes in the circulation and of macrophages at the site of inflammation is about twice as high in the mouse strain C57BL/10 compared with CBA mice, and is probably due to the higher ability of monocyte precursors to respond to the humoral factor by increasing monocyte production (Sluiter, Elzenga-Classen, van der Voort, van der Kley-van Andel & van Furth, 1984). Milon, Gheorghiu, Lagranderie,

Lebestard and Marchal (1984) demonstrated significant differences in the response of mice of strains C57BL/6 and C3H to BCG vaccine injection, with strain C57BL/6 being designated a reactive strain, with an increased number of phagocytes in the blood following the vaccine administration. Kakinuma and Yamamoto (1985) designate the strains C57BL/6 and C57BL/10 as high responders for lung granuloma formation after BCG vaccine injection. The high sensitivity of the C57BL family of strains is the only information at our disposal on the reactivity of our C57BL/10ScSnPh strain. We are aware that even substrains of the same strain may react differently to the same stimulus, e.g. endotoxin (Forni & Coutinho, 1978).

With regard to the effects of inapparent MHV infection, Dempsey, Smith and Morahan (1986) mentioned differences (without specification) among mouse strains. Regardless of these uncertainties, our results suggest that the common action of two

factors, i.e. glucan and the inapparent infection of mice with MHV results in long-term deviations in ferrokinetics and erythropoiesis, resembling the anemia of chronic diseases. Because macrophage parameters appear to be particularly sensitive to MHV infection, the common target of glucan and MHV infection could be the secretory activity of these cells. A permissiveness of some macrophage populations for MHV infection was reported (Bang & Warwick, 1960). Data presented by Boorman *et al.* (1982) and Dempsey, Smith and Morahan (1986) indicate that the number of resident peritoneal macrophages increases after MHV infection.

Taken together, our results demonstrate that inapparent infection with MHV can alter experimental results with immunomodulators. It is tempting to speculate that the coincidence of some infection factors with inflammatory action could lead to clinically important alterations of hemopoiesis, including anemia of chronic diseases.

REFERENCES

- BANG, F. B. & WARWICK, A. (1960). Mouse macrophages as host cells for the mouse hepatitis virus and the genetic basis of their susceptibility. *Proc. natn. Acad. Sci. U.S.A.*, **46**, 1065–1075.
- BIRGEGARD, G. & CARO, J. (1984). Increased ferritin synthesis and iron uptake in inflammatory mouse macrophages. *Scand. J. Haemat.*, **33**, 43–48.
- BONDY, R., KLÍR, P., PŘIBYLOVÁ, M., SVOBODA, T., JELEN, P. & PRAVENEC, M. (1987). Health quality control of laboratory animals of the Czechoslovak Academy of Sciences (CSAV). *Scand. J. Lab. Anim. Sci.*, **14**, 15–24.
- BOORMAN, G. A., LUSTER, M. I., DEAN, J. H., CAMBELL, M. L., LAUER, L. A., TALBY, F. A., WILSON, R. E. & COLLINS, M. J. (1982). Peritoneal macrophage activation caused by naturally accuring mouse hepatitis virus. *Am. J. Path.*, **106**, 110–117.
- BOWERS, G. J., PATCHEN, M. L., MACVITTIE, T. J., HIRSCH, E. F. & FINK, M. P. (1986). A comparative evaluation of particulate and soluble glucan in an endotoxin model. *Int. J. Immunopharmac.*, **8**, 313–321.
- DEGOWIN, R. L. & LASS, S. L. (1985). Chronic inflammation impairs hematopoiesis and survival after irradiation. *J. Lab. clin. Med.*, **105**, 299–304.
- DEMPSEY, W. L., SMITH, A. L. & MORAHAN, P. S. (1986). Effects of inapparent murine hepatitis virus infection on macrophages and host resistance. *J. Leukocyte Biol.*, **39**, 559–565.
- DILUZIO, N. R. (1983). Immunopharmacology of glucan: a broad spectrum enhancer of host defense mechanisms. *Trends pharmac. Sci.*, **4**, 344–347.
- DILUZIO, N. R. (1985). Update on the immunomodulating activities of glucans. *Springer Semin. Immunopath.*, **8**, 387–400.
- FAQUIN, W. C., SCHNEIDER, T. J. & GOLDBERG, M. A. (1992). Effect of inflammatory cytokines on hypoxia-induced erythropoietin production. *Blood*, **79**, 1987–1994.
- FORNI, L. & COUTINHO, A. (1978). An antiserum which recognizes lipopolysaccharide-reactive B cells in the mouse. *Eur. J. Immun.*, **8**, 56–62.
- FRUHMANN, G. J. (1966). Shunting of erythropoiesis in mice following the injection of zymosan. *Life Sci.*, **5**, 1549–1556.
- FRUHMANN, G. J. (1967). Endotoxin-induced shunting of erythropoiesis in mice. *Am. J. Physiol.*, **212**, 1095–1098.
- GALLIN, E. K., GREEN, W. & PATCHEN, M. L. (1992). Comparative effects on particulate and soluble glucan on macrophages of C3H/HeN and C3H/HeJ mice. *Int. J. Immunopharmac.*, **14**, 173–183.
- GRULICH, I. (1977). Osteologische Präparation mit der Verwendung der kadaverikolen Käfer (*Dermestes vulpinus* Fbr., Coleoptera, Insecta). *Lynx (Prague)* n.s., **19**, 87–98.

- HASSID, W. Z., JOSLYN, M. A. & MCCREADY, R. M. (1941). The molecular constitution of an insoluble polysaccharide from yeast *Saccharomyces cerevisiae*. *J. Am. Chem. Soc.*, **63**, 295–298.
- HELLMAN, S. & GRATE, H. (1967). Haematopoietic stem cells: evidence for completing proliferative demands. *Nature*, **216**, 65–66.
- HOLÁ, J., VÁCHA, J. & BOHÁČEK, J. (1979). The preparation of ^{59}Fe -labelled transferrin of high radioactivity for ferrokinetic studies on small laboratory animals. *Acta Haemat.*, **61**, 55–56.
- KAKINUMA, M. & YAMAMOTO, K. (1985). Strain differences in lung granulome formation in response to a BCG cell-wall vaccine in mice. Demonstration of two types of low responders. *Immunology*, **55**, 91–95.
- KLASING, K. C. (1984). Effect of inflammatory agents and interleukin 1 on iron and zinc metabolism. *Am. J. Physiol.*, **247** (Regulatory Integrative Comp. Physiol.), R901–R904.
- LEE, G. R. (1983). The anemia of chronic disease. *Semin. Hemat.*, **20**, 61–80.
- LYNCH, S. R., LIPSCHITZ, D. A., BOTHWELL, T. H. & CHARLTON, R. W. (1974). Iron and the reticuloendothelial system. In *Iron in Biochemistry and Medicine* (eds Jacobs, A. and Worwood, M.), pp. 564–587. Academic Press, London.
- MAURY, C. P. J., ANDERSSON, L. C., TEPPA, A. M., PARTANEN, S. & JUVONEN, E. (1988). Mechanisms of anaemia in rheumatoid arthritis: demonstration of raised interleukin 1 beta concentrations in anaemic patients and of interleukin 1 mediated suppression of normal erythropoiesis and proliferation of human erythroleukaemia (HEL) cells *in vitro*. *Ann. Rheum. Dis.*, **47**, 972–978.
- MCNEILL, T. A. & FLEMING, W. A. (1973). Cellular responsiveness to stimulation *in vitro*: strain differences in hemopoietic colony forming cell responsiveness to stimulating factor and suppression of responsiveness by glucocorticoids. *J. Cell Physiol.*, **82**, 49–58.
- MEANS, R. T. & KRANTZ, S. B. (1992). Progress in understanding the pathogenesis of the anemia of chronic disease. *Blood*, **80**, 1639–1647.
- MILON, G., GHEORGHIU, M., LAGRANDERIE, M., LEBESTARD, M. & MARCHAL, G. (1984). BCG-induces anaemia in mice: no direct effect of the growth of bacilli. *Ann. Immun. (Inst. Pasteur)*, **135C**, 195–204.
- PATCHEN, M. L. (1983). Immunomodulators and hemopoiesis. *Surv. Immun. Res.*, **2**, 237–242.
- PATCHEN, M. L., D'ALESSANDRO, M. M., BROOK, I., BLAKELY, W. F. & MACVITTIE, T. J. (1987). Glucan: mechanisms involved in its "radioprotective" effect. *J. Leukocyte Biol.*, **42**, 95–105.
- PATCHEN, M. L. & MACVITTIE, T. J. (1983). Dose-dependent responses of murine pluripotent stem cells and myeloid and erythroid progenitor cells following administration of the immunomodulating agent glucan. *Immunopharmacology*, **5**, 303–313.
- POSPÍŠIL, M., HOFER, M., PIPALOVÁ, I., VIKLICKÁ, Š., NETÍKOVÁ, J. & ŠANDULA, J. (1992). Enhancement of hematopoietic recovery in gamma-irradiated mice by the joint use of diclofenac, an inhibitor of prostaglandin production, and glucan, a macrophage activator. *Exp. Hemat.*, **20**, 891–895.
- POSPÍŠIL, M., JARÝ, J., NETÍKOVÁ, J. & MAREK, M. (1982). Glucan-induced enhancement of hemopoietic recovery in gamma-irradiated mice. *Experientia*, **38**, 1232–1234.
- POSPÍŠIL, M., TKADLEČEK, L., NETÍKOVÁ, J., PIPALOVÁ, I., VIKLICKÁ, Š., KOZUBÍK, A., VÁCHA, J. & JARÝ, J. (1988). Interstrain differences in the responsiveness of mice to glucan with respect to hematological effects and manifestations of late damage. *Exp. Path.*, **33**, 27–36.
- ROODMAN, G. D. (1987). Mechanism of erythroid suppression in the anemia of chronic disease. *Blood Cells*, **13**, 171–184.
- SHERWOOD, E. R., WILLIAMS, D. L. & DiLUZIO, N. R. (1986). Comparison of the *in vitro* cytolytic effect of hepatic, splenic and peritoneal macrophages from glucan-treated mice on sarcoma M5076. *Meth. Find. Exp. Clin. Pharmac.*, **8**, 157–161.
- SHERWOOD, E. R., WILLIAMS, D. L., McNAMEE, R. B., JONES, E. L., BROWDER, I. W. & DiLUZIO, N. R. (1987). Enhancement of interleukin-1 and interleukin-2 production by soluble glucan. *Int. J. Immunopharmac.*, **9**, 261–267.
- SLUITER, W., ELZENGA-CLASSEN, I., VAN DER VOORT, A., VAN DER KLEY-VAN ANDEL & VAN FURTH, R. (1984). Differences in the response of inbred strains to the factor increasing monocytopenia. *J. exp. Med.*, **159**, 524–536.
- STAATS, J. (1981). Inbred and segregating inbred strains. In *The Mouse in Biomedical Research* (eds Foster, H. L., Small, J. D. and Fox, J. G.), pp. 177–213. Academic Press, New York.
- SUZUKI, M., YOKOTA, K. & SAKAGUCHI, O. (1967). Extraction methods and water solubilization of *Saccharomyces cerevisiae* glucan. *Tohoku Yakka Diagaku Kenkyu Nempo*, **14**, 7–12 (Chem. Abstr., **71**, 109981n, 1969).
- THOMAS, D. B. (1971). The differentiation of transplanted haematopoietic cells derived from bone marrow, spleen and fetal liver. *J. Anat.*, **110**, 297–306.
- VÁCHA, J. (1975). Blood volume in inbred strain BALB/c, CBA/J and C57BL/10 mice determined by means of ^{59}Fe -labelled red cells and ^{59}Fe bound to transferrin. *Physiol. Bohemoslov.*, **24**, 413–419.
- VÁCHA, J., DUNGEL, J. & KLEINWÄCHTER, V. (1978). Determination of heme and non-heme iron content of mouse erythropoietic organs. *Exp. Hemat.*, **6**, 718–724.
- VÁCHA, J., HOLÁ, J., DUNGEL, J. & ZNOJIL, V. (1980). The volume of plasma and erythrocytes in individual bones and in the spleen of mice under physiological conditions and with acute radiation-induced atrophy of the haemopoietic tissue. *Acta Haemat.*, **64**, 165–171.

- VÁCHA, J., MIKESKA, J., ZNOJIL, V., POSPÍŠIL, M., HOLÁ, J., KOZUBÍK, A. & HOŠEK, M. (1988). The fall in the incorporation of labelled iron into the bone marrow during enhanced erythropoiesis: this paradox explained by changes in ferrokinetics and cytokinetics using the model of mice affected by thyroid hormones. *Nucl. Med. Biol.*, **15**, 373–380.
- VÁCHA, J., ZNOJIL, V., HOLÁ, J. & DUNGEL, J. (1982). The internal iron kinetics in mice. *Acta Vet. Brno*, **51**, 3–22.
- VÁCHA, J., ZNOJIL, V., HOLÁ, J. & DUNGEL, J. (1984). Iron kinetics in individual bone marrow regions (bones) in mice. *Acta Vet. Brno*, **53**, 119–136.
- VÁCHA, J., ZNOJIL, V., SEIDEL, H. J. & BARTHEL, E. (1990). Ferrokinetics and erythropoiesis in mice after long-term inhalation of benzene. *Blut*, **60**, 41–47.